PIMPRI CHINCHWAD EDUCATION TRUST's.
PIMPRI CHINCHWAD COLLEGE OF ENGINEERING
(An Autonomous Institute)
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T.Y. B. TECH **Year:** 2025 – 26 **Subject:** FSDL

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Branch: Computer Engineering **Division: B**

Assignment:7

1. Objective

The primary objective of this assignment is to design and develop a web-based **Student Feedback and Review System** using **React.js (frontend)** and **Node.js with Express.js (backend)**, integrated with a **MongoDB database**.

The system aims to provide a dynamic and user-friendly platform where students can:

- Submit feedback for courses, teachers, or institutions
- View feedback summaries and ratings
- Analyze reviews through structured data

Administrators or faculty can:

- Monitor feedback
- Analyze performance trends
- Improve decision-making based on student input

The application demonstrates how modern frontend frameworks and backend technologies can be integrated to build a **full-stack, scalable web application**.

2. Learning Outcomes

After completing this assignment, the following skills and knowledge are gained:

- Understanding **frontend development using React.js**
- Learning **backend development with Node.js and Express.js**
- Designing and implementing **RESTful APIs**
- Working with **MongoDB (NoSQL database)**
- Managing dynamic data (feedback, users, ratings)
- Implementing features like:
    - User authentication (login/register)
    - Feedback submission and management
    - Search and filtering

- Understanding **state management in React**
- Improving **UI/UX design for web applications**
- Enhancing **full-stack integration skills (MERN stack)**

3. Brief Theory

A **Student Feedback System** is a web-based application that allows users (students) to provide structured feedback about courses or faculty. These systems rely on dynamic data handling and efficient backend processing to ensure real-time interaction.

Technologies Used

- **React.js**
  A frontend JavaScript library used to build interactive user interfaces. It enables component-based architecture and efficient rendering.
- **Node.js**
  A runtime environment that allows JavaScript to run on the server side.
- **Express.js**
  A lightweight framework for Node.js that simplifies routing, middleware, and API creation.
- **MongoDB**
  A NoSQL database that stores data in flexible JSON-like documents, ideal for handling varying feedback formats.

4. System Workflow

In this project, the system operates as follows:

- Student and faculty data are stored in **MongoDB**
- Feedback entries include:
  - Student name (or anonymous ID)
  - Course/teacher name
  - Rating (e.g., 1–5)
  - Comments
- The **Express backend** handles API requests and communicates with the database
- The **React frontend** fetches and displays data dynamically

User Actions

- Register/Login into the system
- Submit feedback for courses or faculty
- View existing feedback and ratings
- Search/filter feedback based on criteria

Admin Actions

- View all feedback
- Analyze ratings and trends
- Manage users and feedback data

5. Key Features

- User authentication (JWT-based login system)
- Feedback submission form
- Rating system (stars or numeric)
- Search and filtering functionality

- Dashboard for feedback visualization
- Responsive UI using modern design principles

6. Database Design (MongoDB Collections)

Users Collection

- _id
- name
- email
- password
- role (student/admin)

Feedback Collection

- _id
- studentId
- courseName
- teacherName
- rating
- comments
- createdAt

Code:

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>Student Feedback</title>
  </head>
  <body>
    <div id="root"></div>
    <script type="module" src="/src/main.jsx"></script>
  </body>
</html>
```

```css
@import
url('https://fonts.googleapis.com/css2?family=Inter:wght@400;500;600;700;800&display=swap');

* { box-sizing: border-box; }

:root {
  --bg-grad: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
  --card-bg: rgba(255, 255, 255, 0.95);
  --accent: #6366f1;
  --accent-hover: #4f46e5;
  --accent-grad: linear-gradient(135deg, #6366f1 0%, #8b5cf6 100%);
  --text: #1e293b;
  --muted: #64748b;
  --border: #e2e8f0;
  --error: #ef4444;
  --success: #10b981;
  --shadow-sm: 0 2px 8px rgba(15, 23, 42, 0.06);
  --shadow-md: 0 10px 30px rgba(15, 23, 42, 0.12);
  --shadow-lg: 0 20px 60px rgba(99, 102, 241, 0.25);
}

html, body, #root { height: 100%; }

body {
  font-family: 'Inter', system-ui, sans-serif;
  margin: 0;
  color: var(--text);
  background: var(--bg-grad);
  background-attachment: fixed;
  min-height: 100vh;
  -webkit-font-smoothing: antialiased;
```

```
}

/* Animated background blobs */
body::before, body::after {
  content: '';
  position: fixed;
  border-radius: 50%;
  filter: blur(80px);
  opacity: 0.4;
  z-index: 0;
  pointer-events: none;
  animation: float 20s ease-in-out infinite;
}
body::before {
  width: 500px; height: 500px;
  background: #f472b6;
  top: -100px; left: -100px;
}
body::after {
  width: 400px; height: 400px;
  background: #60a5fa;
  bottom: -100px; right: -100px;
  animation-delay: -10s;
}
@keyframes float {
  0%, 100% { transform: translate(0, 0) scale(1); }
  50% { transform: translate(50px, 30px) scale(1.1); }
}

/* Nav */
nav {
```

```css
  position: relative;
  z-index: 10;
  background: rgba(15, 23, 42, 0.7);
  backdrop-filter: blur(20px);
  -webkit-backdrop-filter: blur(20px);
  color: #fff;
  padding: 16px 32px;
  display: flex;
  gap: 24px;
  align-items: center;
  border-bottom: 1px solid rgba(255, 255, 255, 0.08);
}
nav a {
  color: rgba(255, 255, 255, 0.85);
  text-decoration: none;
  font-weight: 500;
  font-size: 0.95rem;
  transition: color 0.2s;
  position: relative;
}
nav a:hover { color: #fff; }
nav a:first-child {
  font-weight: 800;
  font-size: 1.15rem;
  background: linear-gradient(135deg, #fff 0%, #c7d2fe 100%);
  -webkit-background-clip: text;
  -webkit-text-fill-color: transparent;
  background-clip: text;
}
nav .spacer { flex: 1; }
nav span { color: rgba(255, 255, 255, 0.7); font-size: 0.9rem; }
```

```
/* Container */
.container {
  position: relative;
  z-index: 1;
  max-width: 880px;
  margin: 48px auto;
  padding: 0 20px;
  animation: fadeUp 0.6s ease;
}
@keyframes fadeUp {
  from { opacity: 0; transform: translateY(20px); }
  to { opacity: 1; transform: translateY(0); }
}

/* Cards */
.card {
  background: var(--card-bg);
  backdrop-filter: blur(20px);
  -webkit-backdrop-filter: blur(20px);
  padding: 32px;
  border-radius: 20px;
  box-shadow: var(--shadow-md);
  margin-bottom: 24px;
  border: 1px solid rgba(255, 255, 255, 0.5);
  transition: transform 0.2s, box-shadow 0.2s;
}
.card:hover { box-shadow: var(--shadow-lg); }

.card h2 {
  margin: 0 0 24px;
```

```
  font-size: 1.75rem;
  font-weight: 700;
  background: var(--accent-grad);
  -webkit-background-clip: text;
  -webkit-text-fill-color: transparent;
  background-clip: text;
}
.card h3 { margin: 0 0 16px; font-weight: 600; }

/* Inputs */
input, select, textarea {
  font: inherit;
  padding: 12px 14px;
  border: 1.5px solid var(--border);
  border-radius: 10px;
  width: 100%;
  margin-top: 6px;
  background: #fff;
  transition: border-color 0.2s, box-shadow 0.2s, transform 0.1s;
  color: var(--text);
}
input:focus, select:focus, textarea:focus {
  outline: none;
  border-color: var(--accent);
  box-shadow: 0 0 0 4px rgba(99, 102, 241, 0.12);
}
input[type="range"] { padding: 0; accent-color: var(--accent); }
input[type="checkbox"] { width: auto; accent-color: var(--accent); margin-right: 6px; }

/* Buttons */
button {
```

```
  font: inherit;
  font-weight: 600;
  padding: 11px 22px;
  border: none;
  border-radius: 10px;
  background: var(--accent-grad);
  color: #fff;
  cursor: pointer;
  width: auto;
  transition: transform 0.15s, box-shadow 0.2s, filter 0.2s;
  box-shadow: 0 4px 14px rgba(99, 102, 241, 0.35);
}
button:hover { transform: translateY(-2px); box-shadow: 0 6px 20px rgba(99, 102, 241, 0.5); filter: brightness(1.05); }
button:active { transform: translateY(0); }

nav button {
  background: rgba(255, 255, 255, 0.12);
  box-shadow: none;
  padding: 8px 16px;
  font-size: 0.9rem;
  border: 1px solid rgba(255, 255, 255, 0.2);
}
nav button:hover { background: rgba(255, 255, 255, 0.2); box-shadow: none; }

label {
  display: block;
  margin-top: 18px;
  font-weight: 600;
  font-size: 0.9rem;
  color: var(--text);
```

```css
}

/* Messages */
.error {
  color: var(--error);
  margin-top: 12px;
  padding: 10px 14px;
  background: rgba(239, 68, 68, 0.08);
  border-left: 3px solid var(--error);
  border-radius: 6px;
  font-size: 0.9rem;
}
.success-msg {
  color: var(--success);
  margin-top: 12px;
  padding: 10px 14px;
  background: rgba(16, 185, 129, 0.08);
  border-left: 3px solid var(--success);
  border-radius: 6px;
  font-size: 0.9rem;
}

.row { display: flex; gap: 12px; align-items: center; }

.rating {
  font-size: 1.5rem;
  letter-spacing: 4px;
  background: linear-gradient(135deg, #fbbf24 0%, #f59e0b 100%);
  -webkit-background-clip: text;
  -webkit-text-fill-color: transparent;
  background-clip: text;
```

```
}

/* Tables */
table { width: 100%; border-collapse: collapse; }
th {
  padding: 12px;
  text-align: left;
  font-size: 0.8rem;
  font-weight: 600;
  text-transform: uppercase;
  letter-spacing: 0.05em;
  color: var(--muted);
  border-bottom: 2px solid var(--border);
}
td {
  padding: 14px 12px;
  border-bottom: 1px solid var(--border);
  font-size: 0.95rem;
}
tr:last-child td { border-bottom: none; }
tr:hover td { background: rgba(99, 102, 241, 0.04); }

/* Feedback entry */
.fb-entry {
  padding: 16px 0;
  border-bottom: 1px solid var(--border);
}
.fb-entry:last-child { border-bottom: none; }
.fb-meta { font-size: 0.8rem; color: var(--muted); }
.fb-cats {
  display: inline-flex;
```

```
  gap: 6px;
  flex-wrap: wrap;
  margin-top: 4px;
}
.fb-cats span {
  background: rgba(99, 102, 241, 0.1);
  color: var(--accent);
  padding: 3px 10px;
  border-radius: 12px;
  font-size: 0.75rem;
  font-weight: 500;
}

/* Grid layouts */
.grid-3 {
  display: grid;
  grid-template-columns: 1fr 2fr 1.5fr;
  gap: 14px;
}
@media (max-width: 600px) { .grid-3 { grid-template-columns: 1fr; } }

.course-grid {
  display: grid;
  grid-template-columns: repeat(auto-fill, minmax(220px, 1fr));
  gap: 16px;
}

.course-tile {
  background: linear-gradient(135deg, #ffffff 0%, #f8fafc 100%);
  border: 1.5px solid var(--border);
  border-radius: 16px;
```

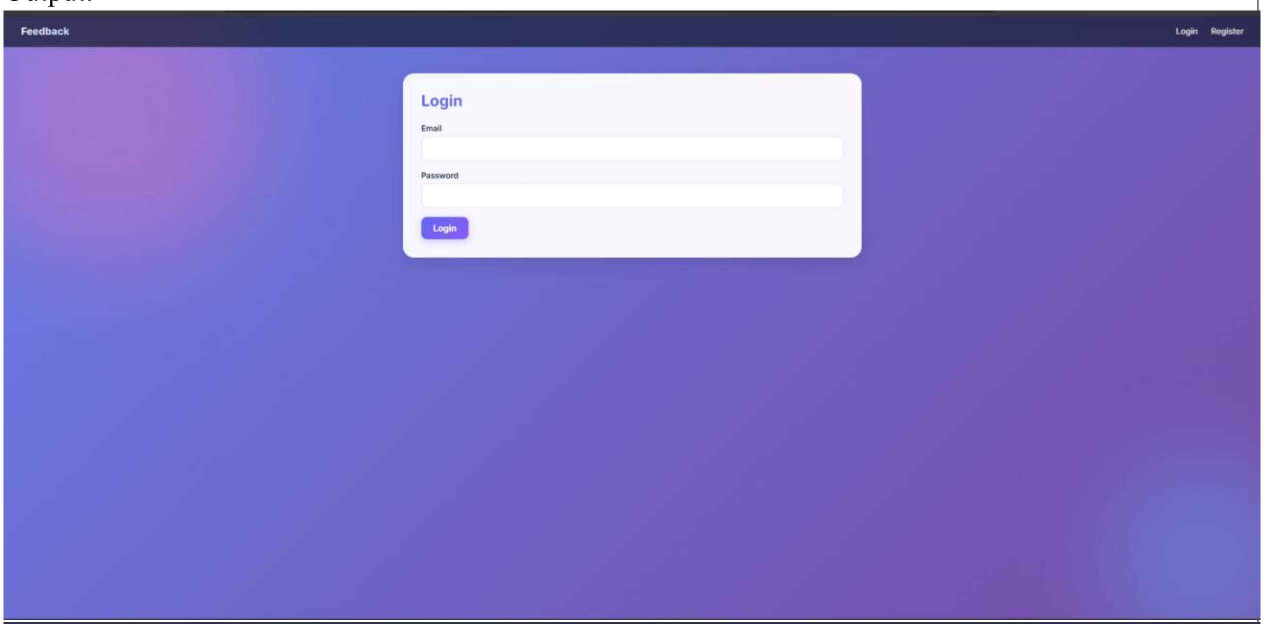
```
  padding: 20px;
  cursor: pointer;
  transition: transform 0.2s, box-shadow 0.2s, border-color 0.2s;
  position: relative;
  overflow: hidden;
}
.course-tile::before {
  content: '';
  position: absolute;
  inset: 0;
  background: var(--accent-grad);
  opacity: 0;
  transition: opacity 0.3s;
  z-index: 0;
}
.course-tile:hover {
  transform: translateY(-4px);
  box-shadow: var(--shadow-lg);
  border-color: var(--accent);
}
.course-tile > * { position: relative; z-index: 1; }

.course-code {
  font-size: 0.75rem;
  font-weight: 700;
  letter-spacing: 0.1em;
  text-transform: uppercase;
  color: var(--accent);
  background: rgba(99, 102, 241, 0.1);
  display: inline-block;
  padding: 3px 10px;
```

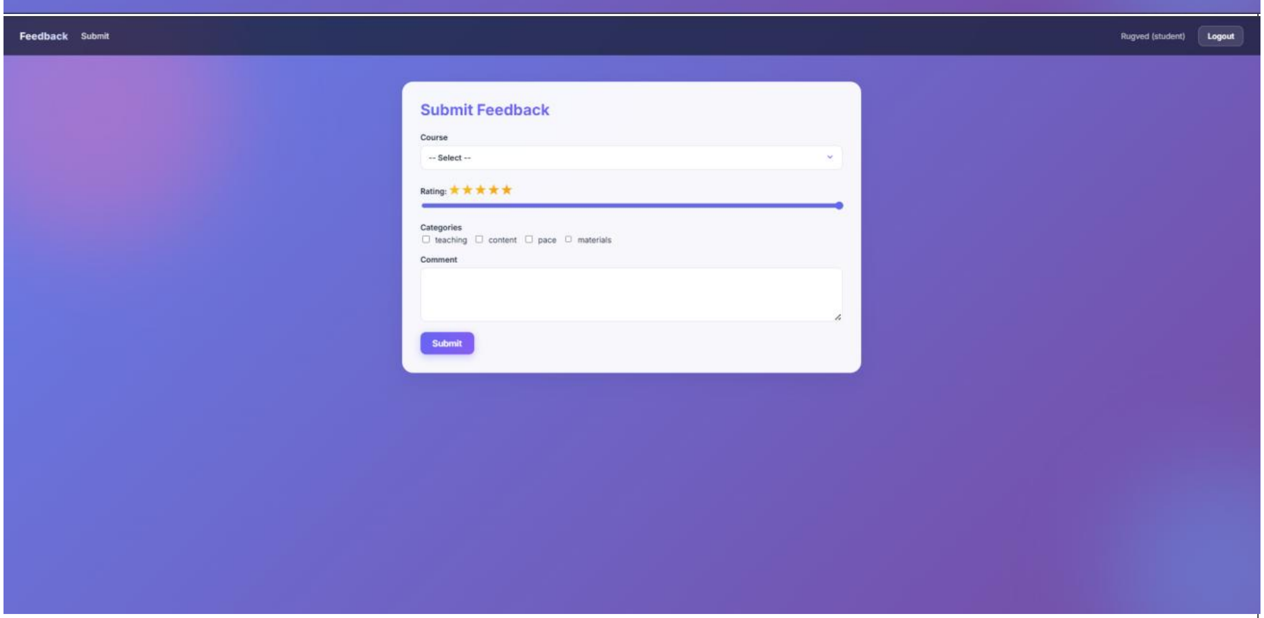
```
  border-radius: 6px;
}
.course-title {
  font-size: 1.1rem;
  font-weight: 700;
  margin-top: 10px;
  color: var(--text);
}
.course-instr {
  font-size: 0.85rem;
  color: var(--muted);
  margin-top: 4px;
}
.course-stats {
  display: flex;
  align-items: center;
  gap: 10px;
  margin-top: 14px;
  padding-top: 14px;
  border-top: 1px dashed var(--border);
  flex-wrap: wrap;
}
.big-rating {
  font-size: 1.6rem;
  font-weight: 800;
  background: var(--accent-grad);
  -webkit-background-clip: text;
  -webkit-text-fill-color: transparent;
  background-clip: text;
}
.count {
```

```
  font-size: 0.8rem;
  color: var(--muted);
  margin-left: auto;
}

.empty {
  text-align: center;
  color: var(--muted);
  padding: 24px;
  font-style: italic;
}

/* Custom select arrow */
select {
  appearance: none;
  -webkit-appearance: none;
  background-image: url("data:image/svg+xml;charset=UTF-8,%3csvg
xmlns='http://www.w3.org/2000/svg' viewBox='0 0 24 24' fill='none' stroke='%236366f1' stroke-
width='2.5' stroke-linecap='round' stroke-linejoin='round'%3e%3cpolyline points='6 9 12 15 18
9'%3e%3c/polyline%3e%3c/svg%3e");
  background-repeat: no-repeat;
  background-position: right 14px center;
  background-size: 16px;
  padding-right: 40px;
}

import axios from 'axios';

const api = axios.create({ baseURL: '/api' });

api.interceptors.request.use((config) => {
  const token = localStorage.getItem('token');
```

```
  if (token) config.headers.Authorization = `Bearer ${token}`;
  return config;
});

export default api;
```

Output:





Conclusion:

Conclusion

This project demonstrates how to build a **full-stack MERN application** by integrating:

- A dynamic **React frontend**
- A scalable **Node.js + Express backend**
- A flexible **MongoDB database**

It highlights real-world application design, including handling user input, managing data efficiently, and building responsive interfaces.

Github Link for all assignments: [Assignments]